

BRACKISH WATER REVERSE OSMOSIS SYSTEM
CAPACITY - 40000GPD

**Techno-Commercial Offer for the Design, Detailed Engineering, Supply,
Erection & Commissioning of Brackish Water RO System**

CLIENT: M/s Al Rawda Farm Mr. SOULIMAN AL REFAI	PROJECT NAME: AL SAMRIYA FACTORY FOR POULTRY PRODUCTION
REF.NO: 26576	DATE: 19/04/2026

1.0 PROJECT & BUSINESS OBJECTIVES:

M/s. Metri Engineering Services W.L.L seeks to provide the lowest and best Techno-commercial for the Supply, Erection & Commissioning of Brackish water RO System.

2.0 DETAILED SCOPE OF WORK:

- Concept development and detailed engineering.
- Design of proposed plant, Detail Engineering.
- Supply and Erection of all Electro-Mechanical equipment's.
- Testing, pre commissioning and commissioning of the plants.
- Submission of drawings such as process flow, process & instrumentation drawing.
- Submission of Operation & Maintenance manual in English.

3.0 DESIGN BASIS:

Capacity	:	40000 Gallons/day
Operating hours	:	24
Maximum Feed flow rate	:	2380GPH
RO Water product rate	:	1667GPH
Reject flow	:	713 GPH
System Recovery	:	70-75%
Power	:	3 phase, 415 V 50Hz
Type of operation	:	Fully Automatic (Except CIP)

Note:

The proposed BWRO system is designed for Max 70-75% recovery at feed water TDS 1100ppm, fluctuation in the feed water quality result changes in system recovery & Water Quality.

4.0 PLANT OPERATING DATA:

Source of Water: Bore Well

The Brackish Water Treatment plant is designed based on the raw bore water parameters analyzed which is furnished below:

S. No	Parameter	Unit	Max Value
1.	pH @ 22°C	-	7.42

2.	Total Dissolved Solids	ppm	1100
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GUARANTEED PRODUCT WATER QUALITY:

In line with input parameters of Brackish water treatment plant will produce following output parameters and shall be guaranteed as below:

S. No	Parameter	Unit	Max Value
1	pH	-	6.5 to 7.5
3	Total Dissolved Solids	ppm	< 50

ABOVE QUALITY AND QUANTITY SHALL BE ACHIEVED PROVIDED:

- ✓ The feed bore water quality is same as per design values provided above.
- ✓ In case of change in feed quality or source, end quality may have variation or changes. Customer will be responsible for this.
- ✓ Fluctuation in the feed water TDS will affect system recovery
- ✓ The operation of system is strictly done as per MES Operation Manual, guidelines and instructions.
- ✓ All log sheets/daily reports are sent to us for monitoring and suggest suitable actions for better and smooth operation of the system each month.
- ✓ Pure chemicals, original spares and consumable specified by us only are used in system.

5.0 CLIENT'S SCOPE:

i. Raw water & treated water storage tanks.

S.NO	DESCRIPTION	VOLUME	QUANTITY
1	Raw Water Storage Tank	As per site design	1 No.
2	Treated Water Storage Tank	As per site design	1 No.

- ii. Raw bore water inlet to feed pump suction line.
- iii. Incoming electrical supply (415V, 50Hz) to the Control panel will be client's scope.
- iv. Reject water drain/transfer line/bore inject will in client's scope.
- v. Coring for pipes and cables, if needed.
- vi. Local equipment mobilization will be in client's scope.
- vii. Shed/store room for the equipment during erection & commissioning will be in client scope.
- viii. Any other equipment or scope which is not included above will be in client's scope.

6.0 CONNECTIVITY:

Inlet Port : 50 mm

Outlet port : 40 mm

7.0 SCOPE OF WORK:

S.NO	Equipment name	Qty
1.	Filter Feed pumps and its accessories	2 Nos (Duty + standby)
2.	Acid chemical dosing system includes pump & tank	1 No.
3.	Manual multimedia filter includes filter media and its accessories	1 No
4.	Antiscalent dosing system includes pump & tank	1 No
5.	5 & 1 micro filter	2 Nos
6.	High pressure pump and its accessories	1 No
7.	RO Skid includes RO housing, RO membrane, dump valve, differential pressure switch, SS pipe work etc...	1 No
8.	Post pH booster dosing system includes pump & tank	1 No
9.	RO flushing & cleaning pump and its accessories	1 No
10.	RO cleaning/flushing tank	1 No
11.	RO cleaning micron filter	1 No
12.	Instruments such as pH meter, TDS meter & ORP meter	1 No
13.	RO control panel	1 No
14.	Documents such as factory test report, warranty certificate, OM manual etc..	1 No
16.	Operator Training	1 No

8.0 PROCESS DESCRIPTION:

Multimedia Filter:

The raw water shall be pumped to Multi media filter for the pre-liminary filtration. Multi Media Filter unit is provided to ensure pre-filtration before the water send to Reverse Osmosis treatment. Raw water shall be filtered by multigrade filter in order to remove coarse materials or Suspended solids which presents in the raw water above 10 Micron size. Multi Media Filter unit is FRP vertical pressure vessel fitted internally with top distributor and Header-lateral type of bottom collecting system. Externally it shall be fitted with UPVC Pipe work & manual backwash butterfly valves. This unit is charged with different sizes of gravels, coarse sand and fine sand anthracite which is supported on different grades of under bed materials. Fine suspended solids, turbidity, silt, free chlorine; Organic particles, pesticides etc., and Giardia cysts, cryptosporidium cysts will be filtered through various grades of sand & carbon bed. The unit is isolated for backwash for every day once.

Anti-scalent Dosing System:

During Reverse Osmosis process, dissolved salts are concentrated on reject side as permeate water is being drawn at specific recovery. As a result of this, dissolved salts concentration increases on reject end and may precipitate if saturation limit exceeds due to any reason. As a resultant effect of salt concentration, scaling (deposition) occurs on membrane surface which results in inferior performance of RO system. For Prevention of scaling a special anti-scalent formulation is dosed in specific proportion. Normally 5% concentrated solution is prepared and required ppm of special anti-scalent is dosed with the help of dosing pump in line. For anti-scalent dosing, one No of dosing pump and one dosing tank in PE construction is provided.

Two-Stage Micron Cartridge Filters:

After chemical dosing, RO feed water may get contaminated with very fine suspended matters. To supply highly filtered water, 2 Nos. of micron cartridge filters at 5 & 1 micron filtration rate are provided. MCF unit is a UPVC housing vertical vessel. Internally it is fitted with PP wound depth type micron cartridge filter element. Water to be filtered passed through this cartridge element to produce highly filtered RO feed water.

RO High Pressure Pumps:

Multi stage vertical centrifugal type High pressure pumps are provided for supply of water to RO system. Each pump shall be fitted with Stainless steel piping. This pump increases pressure required for RO System. Necessary instruments like High & Low pressure switch, pressure gauge & isolation valves are provided for necessary control of system.

RO Module:

Brackish Water RO Unit is provided for producing Total permeates of 40000 Gallons/day flow at 70-75% recovery. RO Unit is comprises of spirally wound Low energy series RO membranes, housed in specially constructed FRP Pressure tubes. FRP pressure tubes are mounted on structural fabricated skid in horizontal modular fashion. Overall construction of skid is expandable modular feature. Feed, product and reject header with u bends and Victaulic couplings are also provided for easy maintenance. For monitoring of various parameters like Conductivity, respective panel mounted indicators / microprocessors based analyzing instruments are provided. An Electrical & Instrumentation panel is provided for automatic operation of RO system and control Philosophy is also provided.

Post pH Adjusting Dosing System:

pH of the product water from RO is high and it is not desirable. To decrease the pH of the product water, pH Correction Dosing Unit has been provided at RO Outlet. A pH meter is provided at the downstream of pH correction system. For this purpose a PE chemical solution tank is provided for preparation of solution. With the help of electrically operated diaphragm type pump, food grade Caustic solution is dosed in the product water line in appropriate proportion.

9.0 TERMINATION POINTS:

S. NO	DESCRIPTION	INTERFACE POINTS
1	Raw Bore Water Inlet	Filter feed pump suction line
2	Product Water Discharge Flange	Max 2 Mtrs from RO Outlet
3	Reject Water Discharge Flange	Max 2 Mtrs from RO Reject

10.0 LISTS OF ELECTROMECHANICAL WITH SPECIFICATIONS:

10.1 FILTER FEED PUMPS WITH AUTOMATIC CHANGEOVER:

Quantity	2 Nos. (1working +1standby)
Flow rate	9m ³ /hr
Discharge head	35m
Type	Horizontal monobloc multistage Centrifugal
Material of housing	Cast Iron
Material of impellor	SS304
Seal Type	Mechanical
Insulation class	F
IP protection	IP55
Motor rating	1.8 kW
Voltage/Frequency	Three phase, 415V, 50Hz
Accessories	Base frame, isolation butterfly valves, suction and discharge header, NRV, pressure gauges etc...
Make	Grundfos- India

10.2 MULTIMEDIA FILTERS:

Quantity	1 No.
Flow rate	9m ³ /hr
Type	Cylindrical Vertical
Diameter	30inches
Height	72inches
Material of housing	Fiber glass reinforced polymer
Max Working Pressure	10 bar
Filtration Media	Gravels, Coarse, Fine Sands & Anthracite
Control Valve	Manual butterfly valve
Accessories	Top & bottom hub lateral distributor, air bleed valve, pressure gauges etc...
Make	Pentair- India

10.3 ANTI SCALENT DOSING SYSTEM:

Quantity	1 No
Flow rate	0 – 5 LPH



Delivery head	50 m
Type	Solenoid
MOC of housing	Poly propylene
Stroke adjustment	Manual
Motor Rating	12w
Dosing tank	Volume: 100 Liters, Material: Poly Ethylene
Voltage/Frequency	Single phase, 230V, 50Hz
Accessories	Foot valve, suction strainer, NRV etc...
Make	Etatron- Italy

10.4 HCL ACID DOSING SYSTEM:

Quantity	1 No
Flow rate	0 – 5 LPH
Delivery head	50 m
Type	Solenoid
MOC of housing	Poly propylene
Stroke adjustment	Manual
Motor Rating	12w
Dosing tank	Volume: 100 Liters, Material: Poly Ethylene
Voltage/Frequency	Single phase, 230V, 50Hz
Accessories	Foot valve, suction strainer, NRV etc...
Make	Etatron- Italy

10.5 MICRON CARTRIDGE FILTERS:

Quantity	2 Nos.
Type	Cylindrical Vertical
MOC of housing	UPVC
Degree of filtration	5 & 1 micron
Type of cartridge	Spun wound (food grade)
Accessories	Drain valve, Air release valve, manual butterfly valve etc.,
Make	MES

10.6 HIGH PRESSURE PUMP:

Quantity	1 No
Flow rate	9m ³ /hr
Discharge head	120m
Type	Horizontal monobloc multistage Centrifugal
Material of housing	Cast iron
Material of impellor	SS304
Seal Type	Mechanical
Insulation class	F
IP protection	IP55
Motor rating	7.5 kW
Voltage/Frequency	Three phase, 415V, 50Hz
Accessories	Base frame, isolation butterfly valves, suction and discharge header, NRV, pressure gauges etc...
Make	Grundfos- India

10.7 RO MODULE:

Quantity	1 Set
System recovery	70-75%
Feed water TDS	1100ppm
Product water TDS	50ppm
Reject water TDS	4200ppm
Housing Material	Rugged fiber glass
Size of housing	8 inch, 4 element long
No of housing	2
Pressure rating of housing	300 psi
Membrane type	Thin film TFC
MOC of membrane	Polyamide
No of membranes	8
Make of housing	Pentair Ecoline
Accessories	End cap, thrust cone, brine seal, End cap adaptor, interconnector etc..
Make of membrane	Filmtec -USA

10.8 POST pH BOOSTER DOSING SYSTEM:

Quantity	1 No
Flow rate	0 – 5 LPH
Delivery head	50 m
Type	Solenoid
MOC of housing	Poly propylene
Stroke adjustment	Manual
Motor Rating	12w
Dosing tank	Volume: 100 Liters, Material: Poly Ethylene
Voltage/Frequency	Single phase, 230V, 50Hz
Accessories	Foot valve, suction strainer, NRV etc...
Make	Etatron- Italy

RO MEMBRANE MAINTENANCE CLEANING SYSTEM

10.9 RO CLEANING PUMP:

Quantity	1 No
Flow rate	12m3/hr
Discharge head	35m
Type	Horizontal monobloc multistage Centrifugal
Material of housing	Cast Iron
Material of impellor	SS316
Seal Type	Mechanical
Insulation class	F
IP protection	IP55
Motor rating	2.0 kW
Voltage/Frequency	Three phase, 415V, 50Hz
Accessories	Base frame, isolation butterfly valves, suction and discharge header, NRV, pressure gauges etc...
Make	Grundfos- India

10.10 RO CHEMICAL CLEANING STORAGE TANK:

Quantity	1 No.
Type	Cylindrical Vertical
MOC	LDPE
Volume	100 Gallons



Accessories	Over flow piping, high level switch
Make	Almuftah – Qatar

10.11 RO CLEANING MICRON CARTRIDGE FILTER:

Quantity	1 No
Type	Cylindrical Vertical
MOC of housing	UPVC
Degree of filtration	5 micron
Type of cartridge	Spun wound
Housing material	UPVC
Cartridge dimension	2.5 inch dia, 30 inch long
Max operating pressure	3.5 bar
Accessories	Drain valve, Air release valve etc.,
Make	MES

10.12 INSTRUMENTS:

Quantity	1 Lot
Rota meter	Location: RO Feed, RO Reject. Make: Aster
Pressure gauges	Location: All pumps, Sand Filters, Cartridge filter, RO in & Out. Make: Baumer
High pressure Switch & Low pressure Switch	Location: High pressure pump Inlet & Outlet. Make: Danfoss
Conductivity Meter	Location: RO Product. Make: Create
Level switches	Location: Filter feed tanks & Treated water tanks. Make: Mac
pH Meter	Location: RO Feed & RO permeate. Make: Create

10.13 PIPES & FITTINGS FOR WATER LINE:

Quantity	1 Lot
MOC	UPVC-Class E & Flexible reinforced plastic polyethylene.
Accessories	Flanges, Tee joints, elbow, isolation valves, foot valves etc.,
Make	Astral/Atlas

10.14 PIPES & FITTINGS FOR RO LINE:

Quantity	1 Lot
MOC	SS
Accessories	Flanges, Tee joints, elbow, isolation valves, foot valves etc.,
Make	MES

10.15 RO ELECTRICAL PANEL:

Quantity	1 No.
Type	Skid mounted- Auto/Manual mode & safety instruments to ensure that the equipment and accessories are protected.
Controls	• 1 Low & high pressure cut-off safety switch • 1 Unit running indicator lamp • 1 On & Off switch • 1 Selector switch for operating the plant in automatic or manual mode. • Level switches for feed & product line.



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Panel protection	IP 55
Make of components	ABB/L &T/Siemens/Schneider

PRICE DETAILS WITH COMMERCIAL TERMS & CONDITIONS:

SR. NO	DESCRIPTION	QTY	RATE QR	AMOUNT QR
1	Supply, Erection & Commissioning of Brackish Water Treatment Plant – 40000 Gallons/day as per specification mentioned above.	1 Unit	170,000.00	170,000.00
	Net-Amount QAR			170,000.00

1) Prices:

The above price is for our standard design and scope of work indicated in the attached technical section should there be any change in the same the final prices shall be agreed upon by discussions / negotiations.

2) Payment Terms:

The mode of payment and the milestones for payment shall be as under:

40% Advance along with fully defined and accepted purchase order

50% Against Delivery

10% - Against Completion

3) Delivery:

We will be in a position to deliver the equipment within 8-12 weeks time from the date of receipt of Down Payment with LPO.

4) Validity of Offer:

This offer shall remain valid for a period of 30 days from the date of submission.

5) Acceptance of Order:

MES will accept the order on the basis of agreed terms of payment and conditions of contract.

6) Warranty:

Plant will be subject to warranty of 12 months from the date of testing. This warranty will be conditional upon proper maintenance and service work being undertaken and recorded in accordance with MES operation and maintenance procedure.

7) Statutory Variation:

Any change in Govt., duties levies, taxes or policies having direct effect on our above quoted prices will be at customers account for which documentary evidence will be submitted on request.

We hope our offer will fulfill your requirements and we expecting your valuable order as soon. For anymore clarifications please feel free to contact us.

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